SUDIPTA BANIK

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SUMMARY

Software Engineer with almost 7 years of experience at Uber, Grab, and FourKites specializing in scalable, fault-tolerant distributed systems and event-driven backends. Proficient in Go, Java, SQL, Redis, and Kafka. Drove backend projects end to end, from design to production, across systems handling millions of transactions and jobs daily, spanning regulated fintech, distributed compute, and global logistics domains. Open to relocation.

WORK EXPERIENCE

UBER

Software Engineer II — Oct 2024 – Present
Batch Compute Team — Bangalore, India

- **Developed an automated resilience mechanism** for internal Apache YARN cluster federation using circuit breakers and adaptive retries, ensuring **system stability** and service recovery in **under 60 seconds** during downstream failures.
- **Implemented a tiered workload protection component in Kubernetes** to safeguard **business-critical processes**, using real-time configMap driven policy enforcement to manage system outages and guarantee resources for **high priority operations**.
- **Improved internal data platform efficiency** by re-engineering job submission logic, **eliminating close to 20-second scheduling delays** for batch data processing workloads.
- **Built a centralized failover controller** to maintain continuous **platform availability**, enabling rapid apache yarn cluster isolation to prevent service interruptions during large-scale network outages.
- Enhanced Uber's internal Kubernetes batch job observability dashboard with workload-type and job-ID search, and a resource-usage heatmap - **Improving on-call debuggability** and reducing time-to-insight during incidents.

GRAB

Senior Software Engineer — Oct 2023 – Oct 2024
Lending Core Team, Fintech — Bangalore, India

- Redesigned the loan offer and credit limit generation from a monthly batch job to **real-time event-driven processing using Kafka**, reducing peak DB load by ~30% and reducing loan offer waiting time for **millions of drivers**.
- Reduced loan creation API p95 latency by almost 32% via **MySQL batch inserts** for instalment. The reduction in network round-trips and lock acquisitions **improved throughput during high-concurrency database writes**.
- Developed a **sub-second data aggregator API** integrating 5+ internal APIs for banking partners; with **partial response fallback for high availability**.
- **Improved the internal Kafka consumer framework reliability** via graceful shutdown re-queuing, reducing message loss to near zero across financial transaction pipelines.
- **Mentored 2 new engineers through onboarding**, maintaining team velocity during hiring season.

Software Engineer — Oct 2021 – Sep 2023
Lending PayLater Team, Fintech — Bangalore, India

- For Grab's Indonesia PayLater launch, **designed and scaled** the **Refund API** to handle 10% of the country's user base. Used **idempotent and state machine design patterns**, automating retries, error recovery and complex refund flows, which handled 3M+ users gracefully.
- Built a **tiered service fee module for real-time charge APIs and batch billing systems**, implementing slab-rate logic per Indonesian regulatory requirements.
- Designed a configurable credit risk assessment module, **collaborating with product teams and external credit bureaus** to integrate country-specific requirements across **multiple Southeast Asian markets**.
- Developed a **robust lending credit score API**, integrating credit score from data-science team models and user metadata services, ensuring **adherence to REST API design and security best practices**.

- Improved CI/CD build speeds by ~18% and increased unit test coverage by ~35% through Go version upgrade, glide to go mod adoption, and systematic refactoring following **clean code adherence and dependency inversions**.

FOURKITES INC.

Software Engineer Jun 2019 – Oct 2021
Multimodal Supply Chain Visibility Team Chennai, India

- Architected a **Redis-backed caching layer to enable direct integration** with 10+ global maritime carriers, reducing reliance on costly third-party data providers. Designed carrier-specific configs to comply with **API rate limits and built end-to-end data parsers and update flows**, which was later adopted by rail and air teams.
- **Mentored an intern** on feature implementation and code review process for maritime integration module, accelerating knowledge transfer and team productivity.
- Improved **global address search API availability** by replacing a shared Elasticsearch dependency with a dedicated Port Autocomplete API utilising **composite SQL indexing and in-memory cache** for 80,000+ ports and achieved **sub-100ms latency**.
- **Designed an async Kafka-based maritime event enrichment pipeline** for ETA/ETD updates by **integrating data from multiple internal microservices** into a unified callback payload — ensuring event persistence, configurability, and reliable real-time shipment update delivery to customers.
- Eliminated error-prone QA processes by replacing SSH-based Ruby script execution with an **internal shipment event simulation tool** with dropdown-driven UI, event-specific data fields, and background worker integration. Delegated some tasks to an intern for their learning.
- Added ocean-specific features to the shipment cloning module, enabling **domain-specific data replication that improved accuracy and impact** of new client sales demos.

EDUCATION

Jadavpur University
Aug 2015 – Aug 2019
Bachelor of Engineering *in Electrical Engineering* Kolkata, India
Result: CGPA: 7.91/10

Relevant Coursework: Principles of Software Engineering, Communication Engineering and Computer Networks, Sequential Systems and Microprocessors, Computer Fundamentals, Basics of Numerical Methods and Programming, Computer Programming Laboratory, Introduction to Statistical and Probabilistic Methods, Engineering Mathematics.

TECHNICAL SKILLS

- **Languages:** Golang, Java, Python, SQL, C++
- **Backend & API:** REST, gRPC, Microservices, Event-Driven Architecture, Spring Boot
- **Messaging & Infrastructure:** Apache Kafka, Kubernetes, Docker, Apache YARN, Linux
- **Databases & Caching:** MySQL, PostgreSQL, Cassandra, Redis, Aerospike
- **Cloud & Observability:** AWS, GCP, Datadog, Git

ACHIEVEMENTS

- **The Grab Way Award (Q1 2023)** — Recognized for leading engineering improvements in the Buy Now Pay Later (BNPL) project, enhancing system reliability and supporting business growth.

PROJECTS

Travelling Thief Problem *2018*
Explored metaheuristic approaches to approximate solutions for the Travelling Thief Problem, a composite NP-hard problem coupling TSP and 0/1 knapsack selection implemented in MATLAB, under the guidance of professor at Indian Statistical Institute, Kolkata. *Source: github.com/baniksudipta/ttp_project*

LANGUAGES

English: Full Professional Proficiency **Bengali:** Native